Deep Generative Models II

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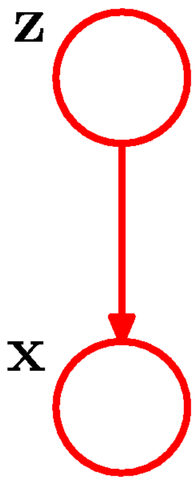
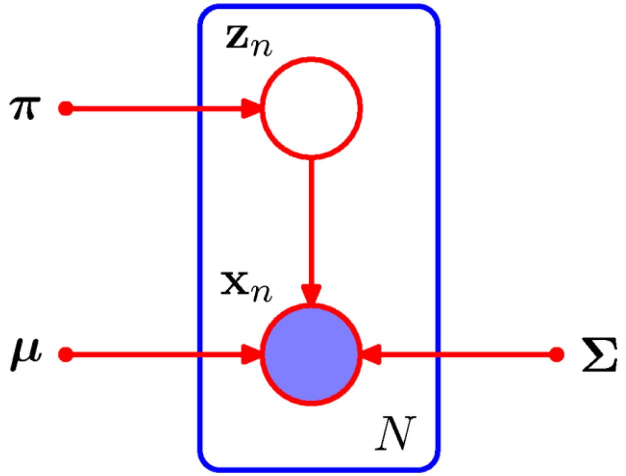
Outline

- Auto-encoding Variational Bayes
- Generative Adversarial Networks

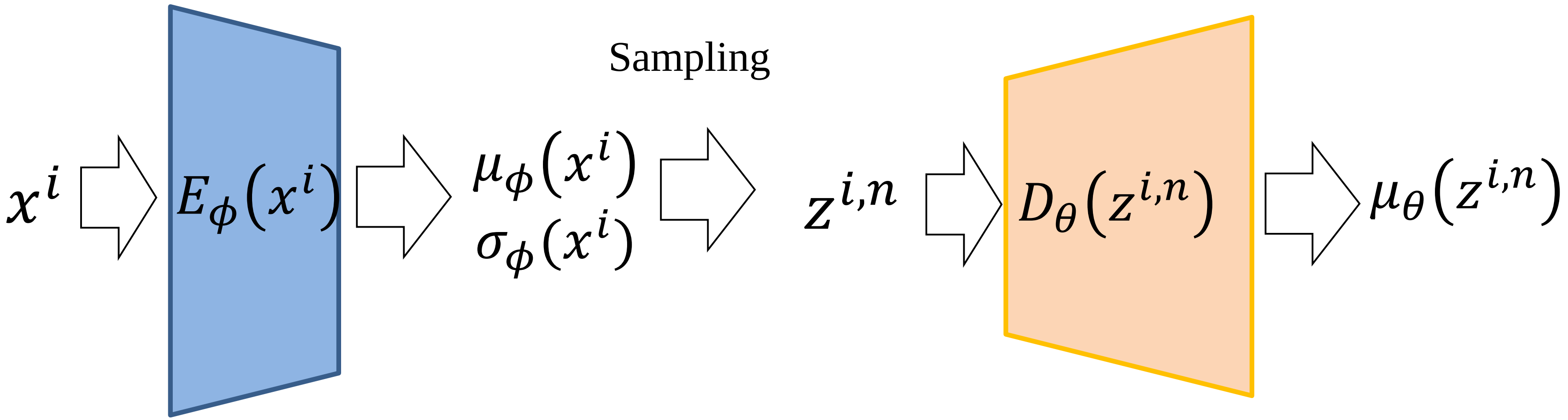
Latent Variables and Probability Distribution





VAE revisited

$$L(\theta,\phi) = -KL\big(q_\phi(z|x)\big|p_\theta(z)\big) + \boxed{\frac{1}{N}\sum_n \log p_\theta(x|z^n)}$$



For loss, we can simply use MSE + KL-constraint term:

$$\frac{1}{N}\sum_n |D_\theta(z^{i,n}) - x^i|^2 + \frac{1}{2}\sum_{j=1}^{J}(1 + \log((\sigma_j{}^2)) - (\mu_j)^2 - (\sigma_j{}^2)$$

VAE revisited

Now we have:

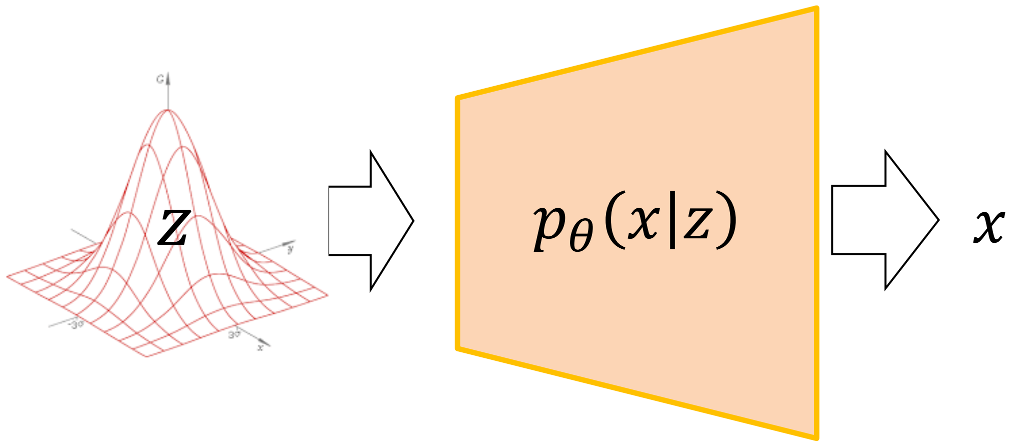
1) likelihood $p_\theta(x|z)$ (Decoder)
2) posterior $p_\theta(z|x) \cong q_\phi(z|x)$ (Encoder)
3) prior $p_\theta(z)$ (by assumption)

- Bayesian decision

- Generative model

$$x \sim p_\theta(x) = \int p_\theta(x|z)p(z)dz$$

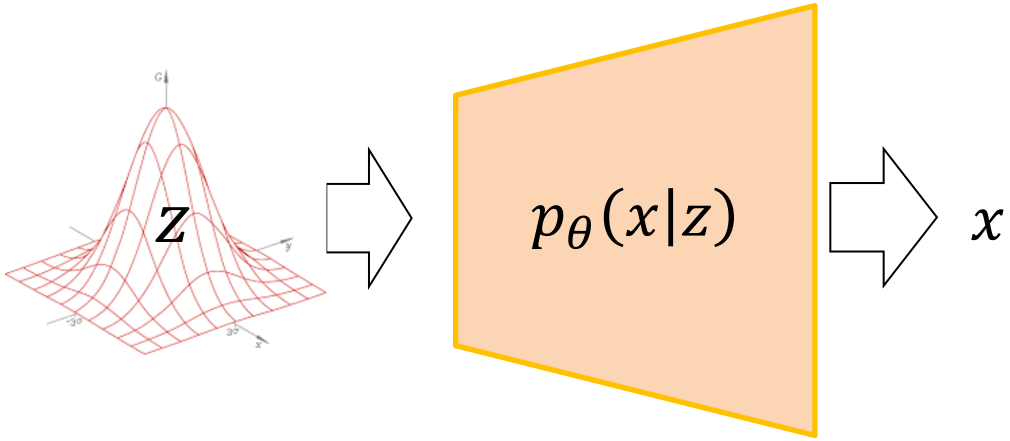
$$x \sim p_\theta(x|z) \text{ where } z \sim p_\theta(z)$$



Generative Adversarial Networks

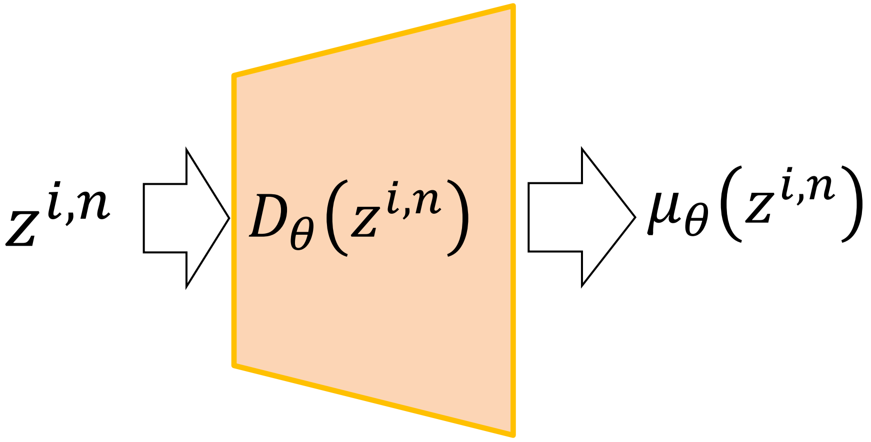
In the aspect of Generative Model:

VAE: $x \sim p_\theta(x|z)$ where $z \sim p_\theta(z)$



How do we directly obtain $\mathrm{x} = f(z)$??

(where $z \sim p_\theta(z)$)



Generative Adversarial Networks

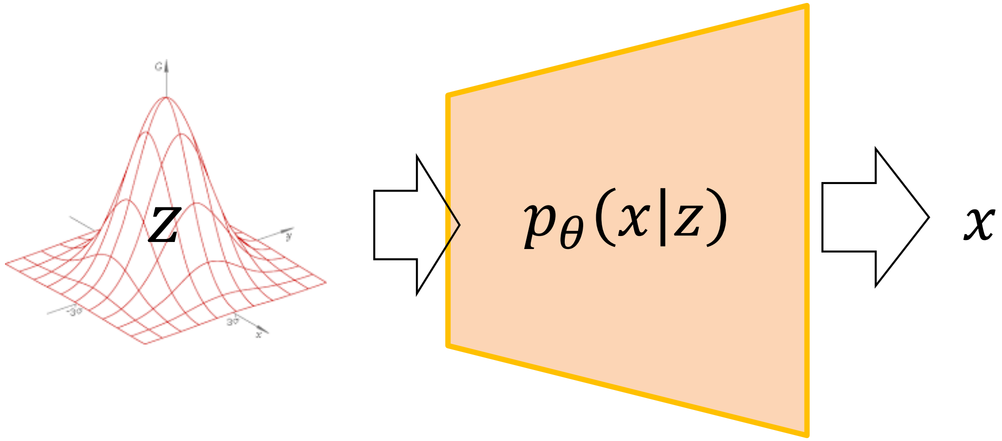
It is possible to find $f(z)$ if:

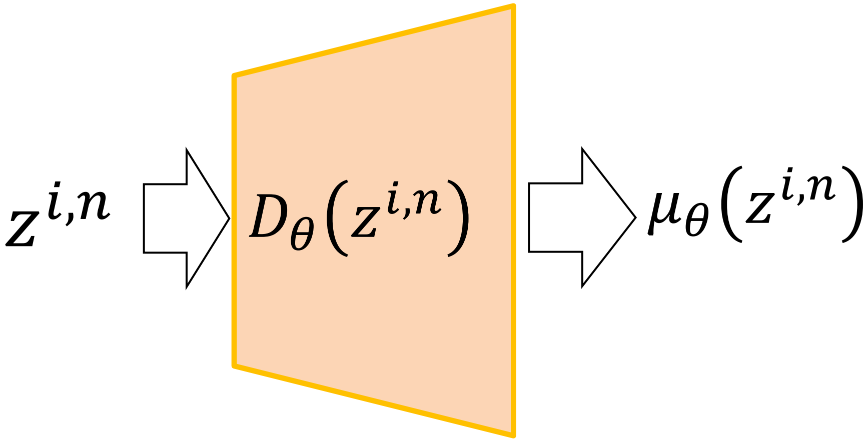
we have pairs of z and x (supervised learning),
which is almost impossible assumption in our case

the only thing that we have is:

$$x_{data}$$

which implicitly represents $p(x_{data})$



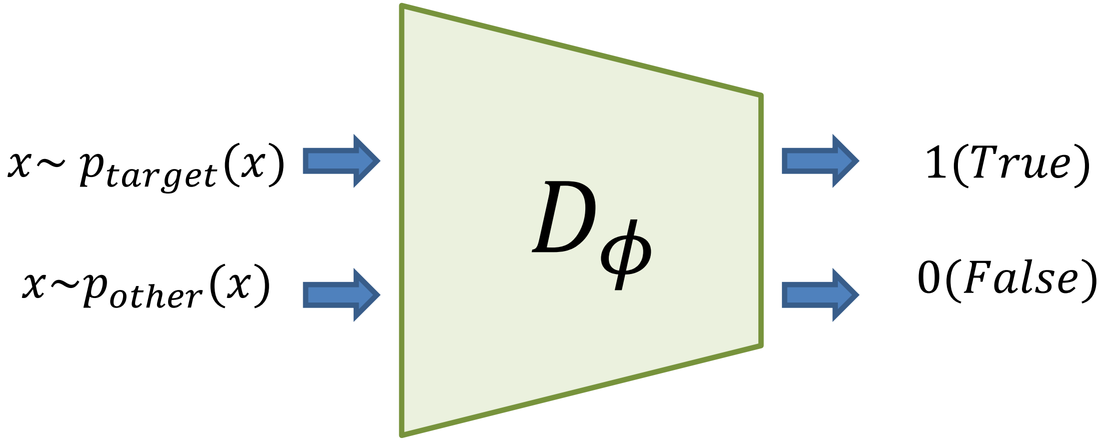


Discriminator

Key idea of Generative Adversarial Networks

1) Domain **D**iscriminator

- assume that we have a perfect discriminator:

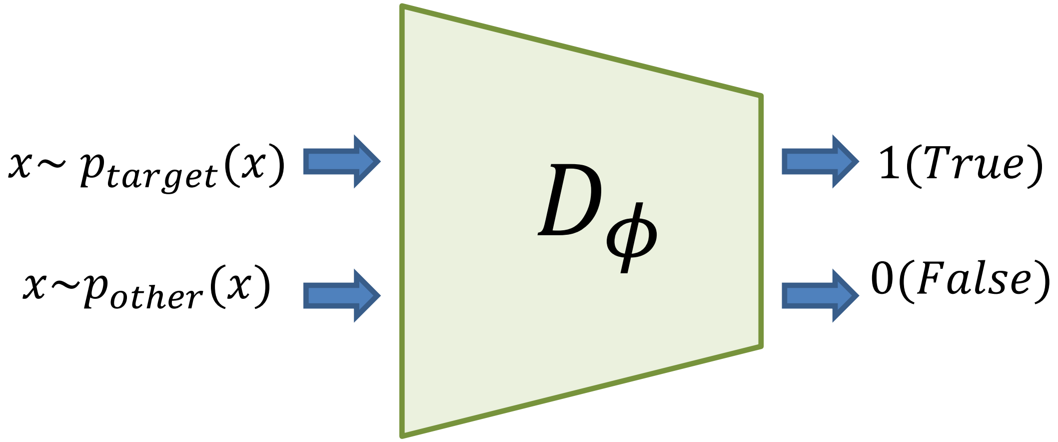


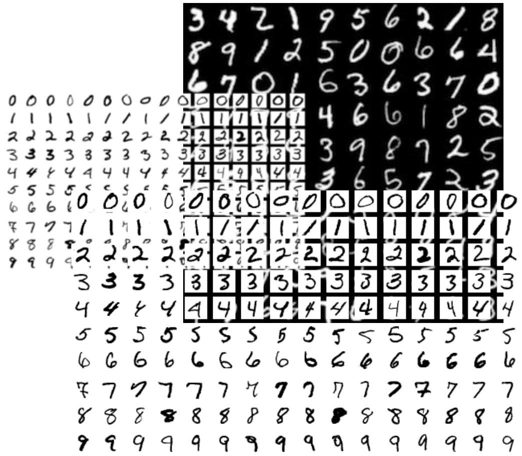
Discriminator

Key idea of Generative Adversarial Networks

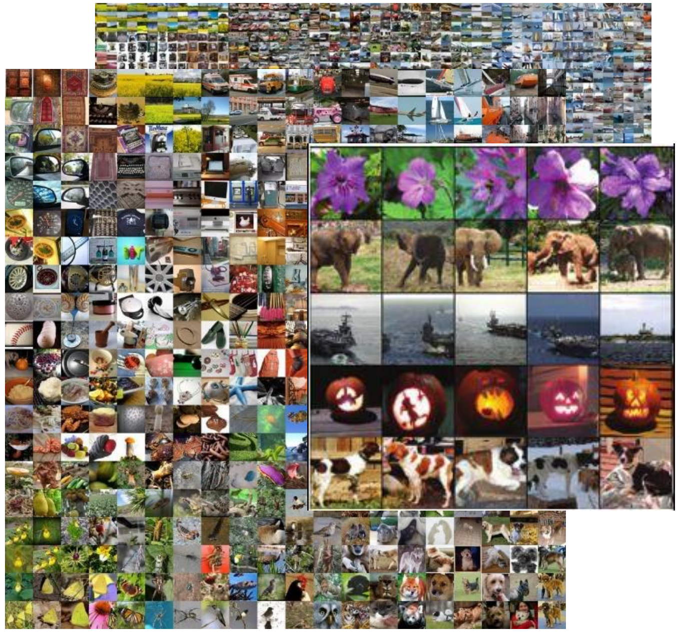
1) Domain Discriminator

- assume that we have a perfect discriminator, which can classifies the target domain and the other domain:





$x \sim p_{target}(x)$



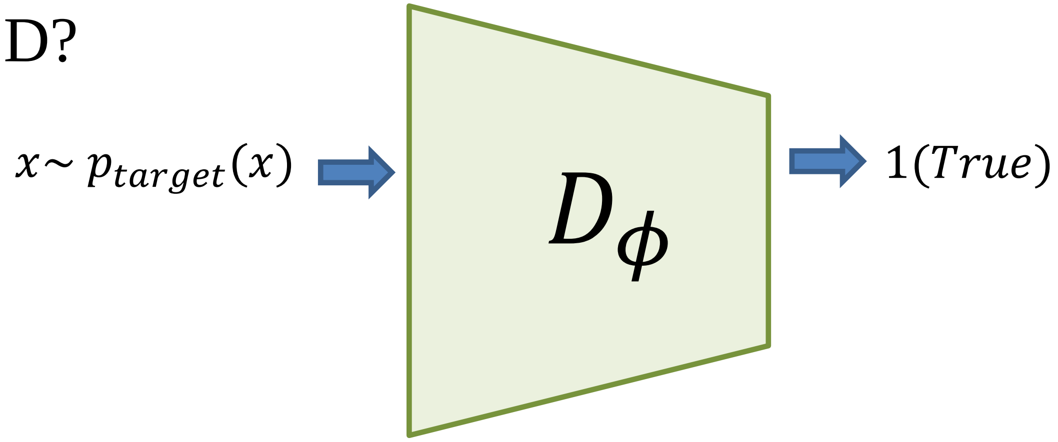
$x \sim p_{other}(x)$

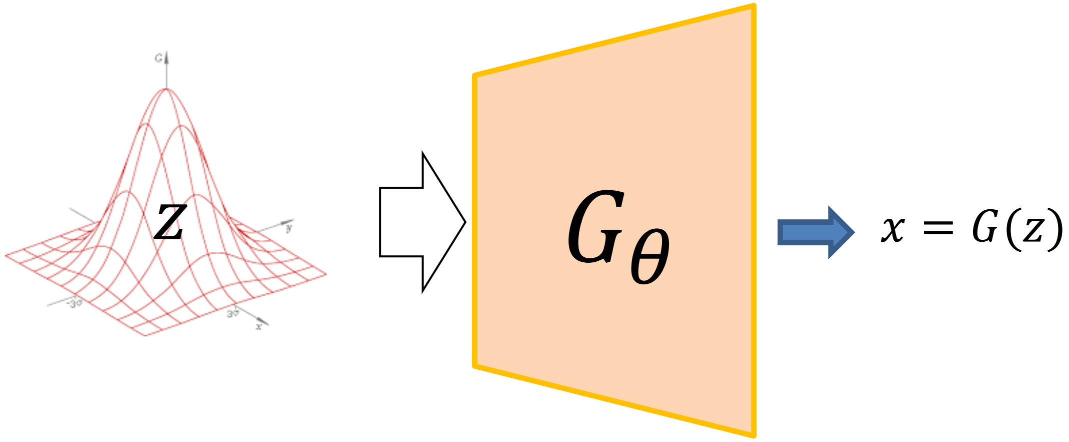
Generator

Key idea of Generative Adversarial Networks

2) Generator

- what if we have G which can deceive D?





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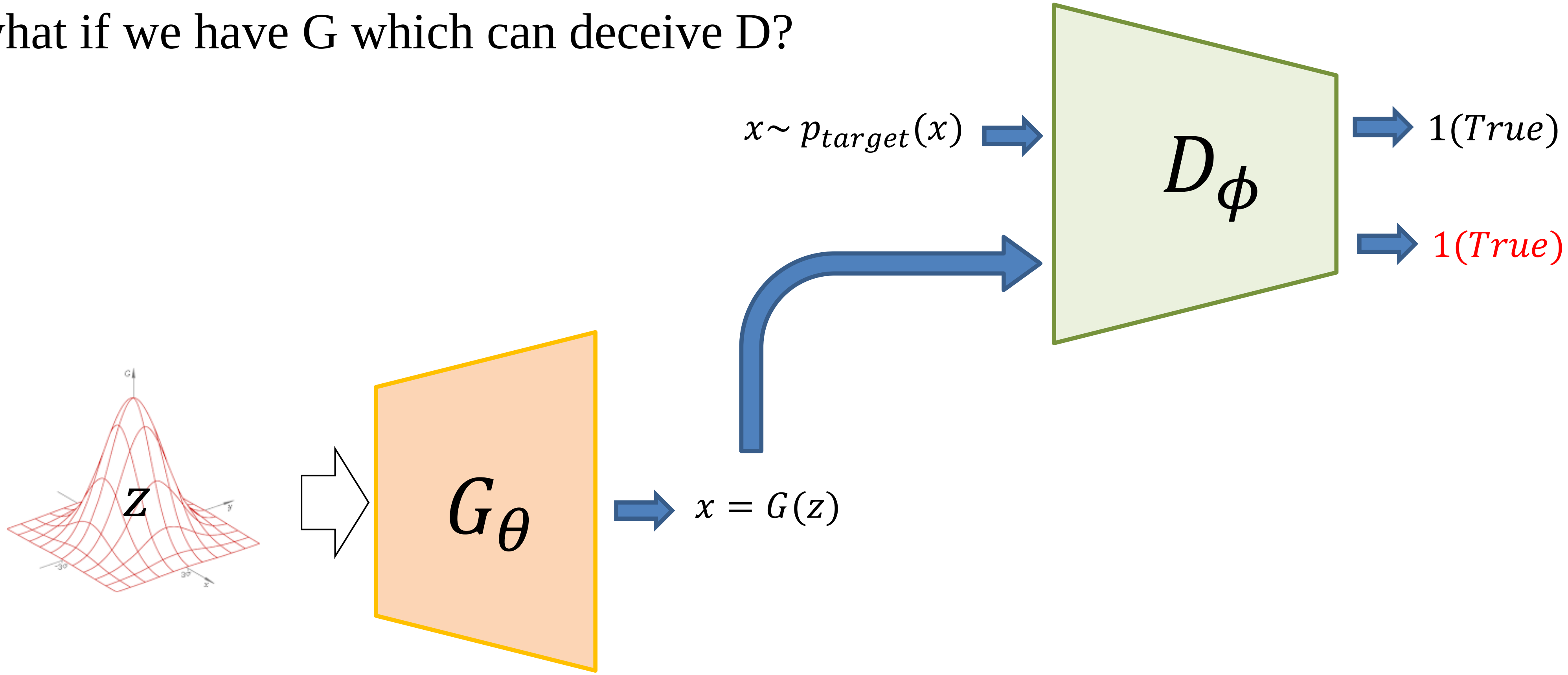

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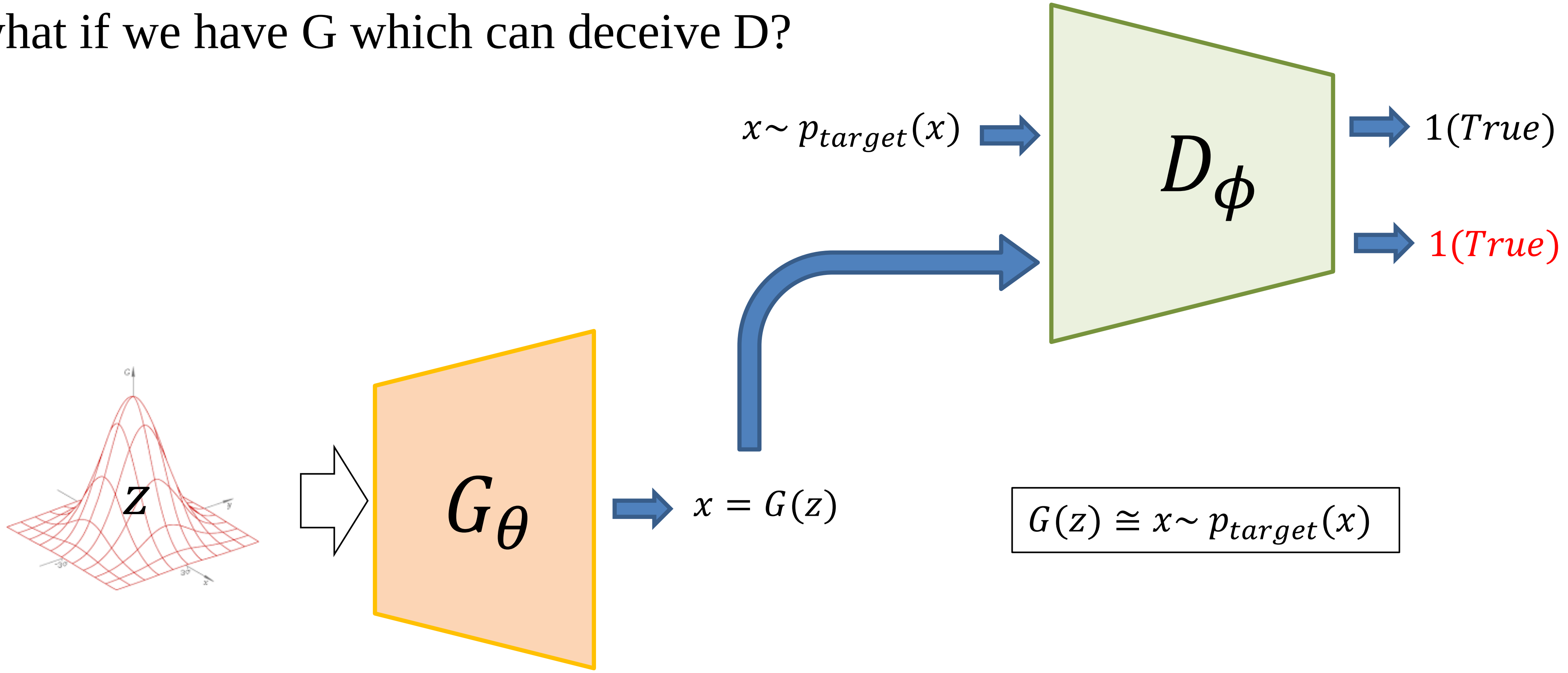

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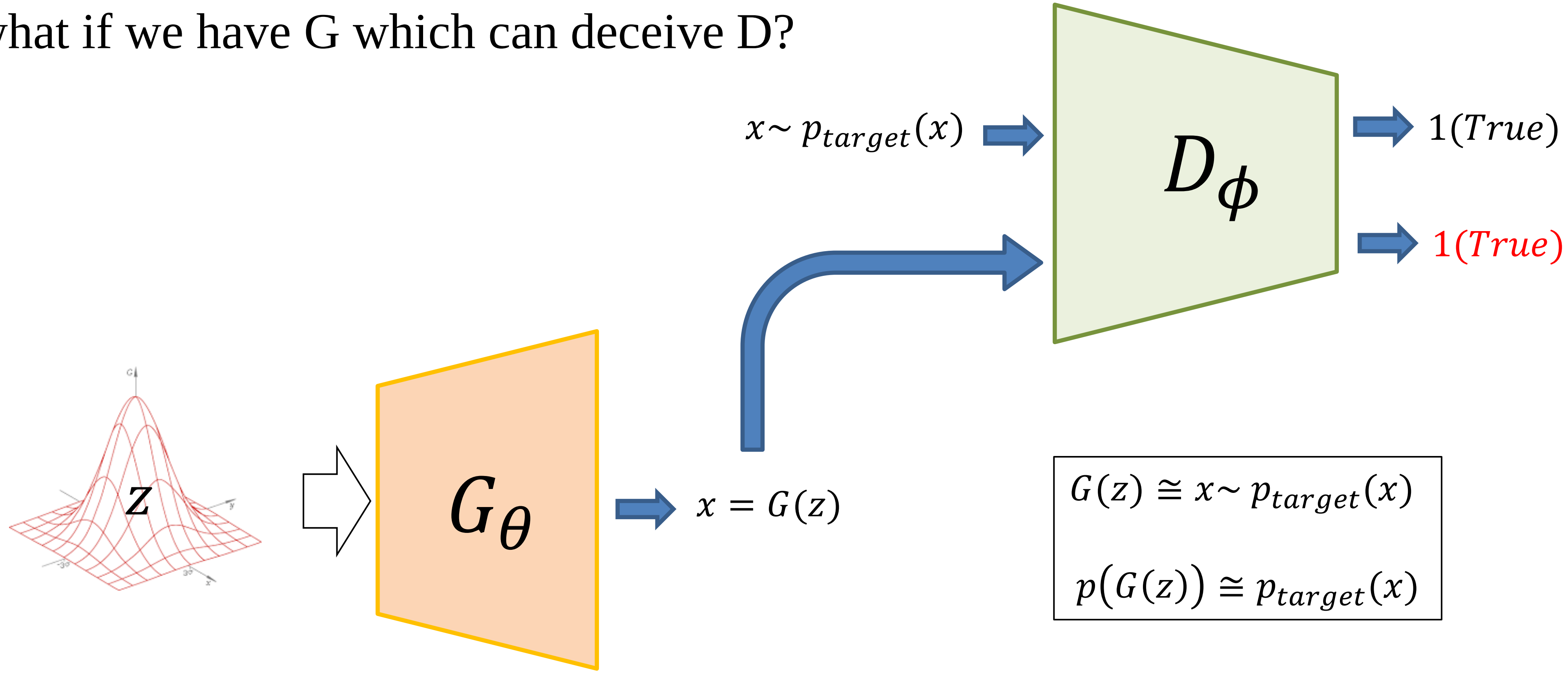


Loss function of GAN

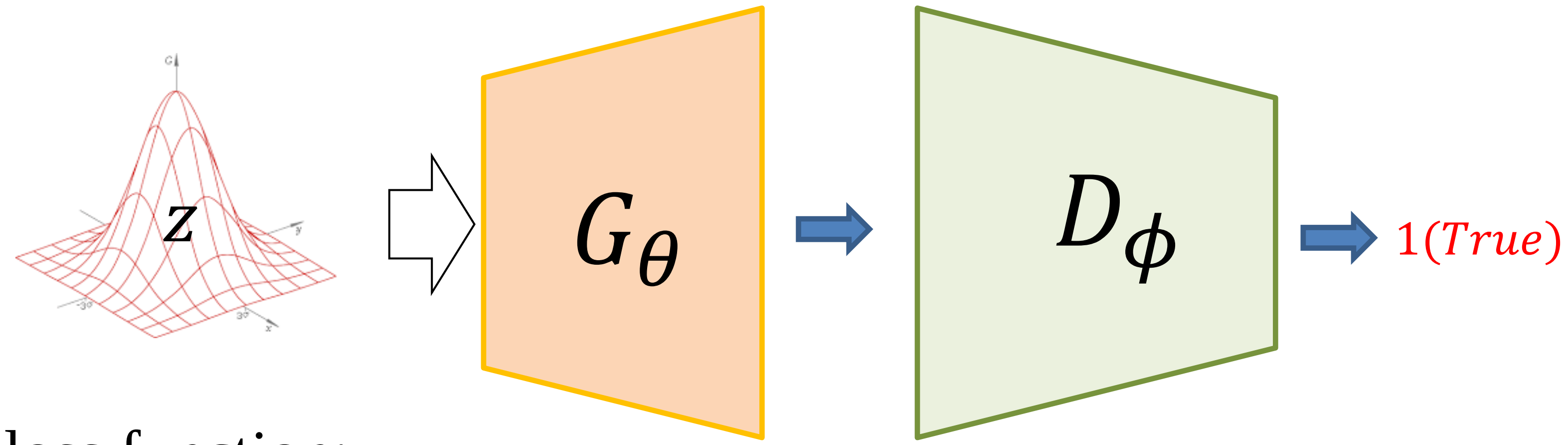
Key idea of Generative Adversarial Networks

2) Generator

- what if we have G which can deceive D?

$$G(z) \cong x \sim p_{target}(x)$$

$$p\big(G(z)\big) \cong p_{target}(x)$$



target loss function:

$$loss = 1 - D_\phi\big(G_\theta(z)\big)$$

Loss function of GAN

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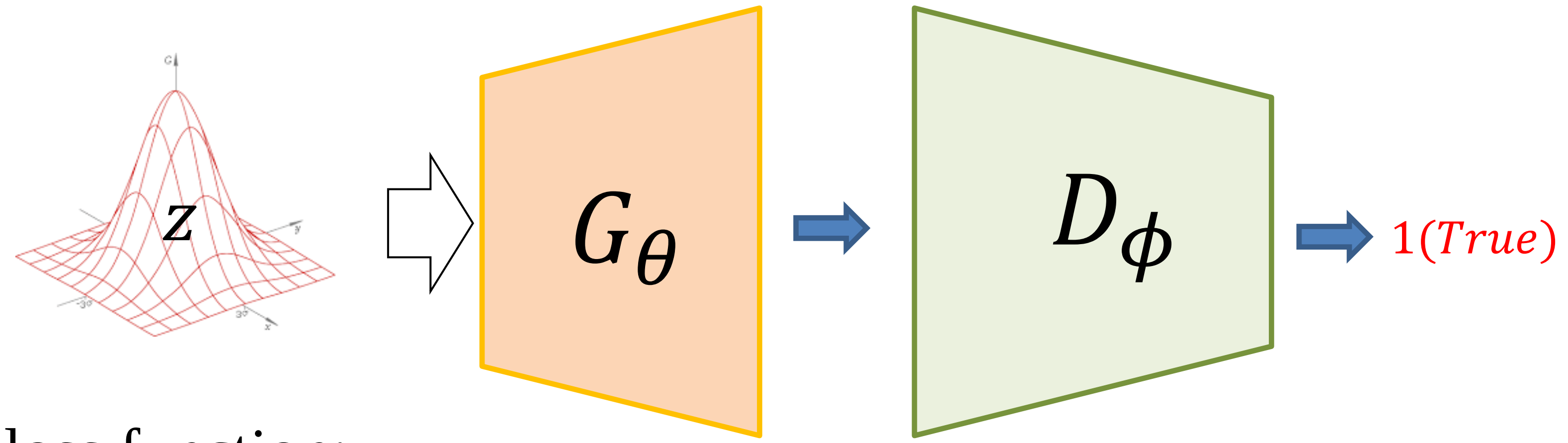

target loss function:

$$loss = \frac{1}{N}\sum_{z \sim p(z)} 1 - D_\phi\big(G_\theta(z)\big)$$

Loss function of GAN

Key idea of Generative Adversarial Networks
2) Generator
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$$G(z) \cong x \sim p_{target}(x)$$

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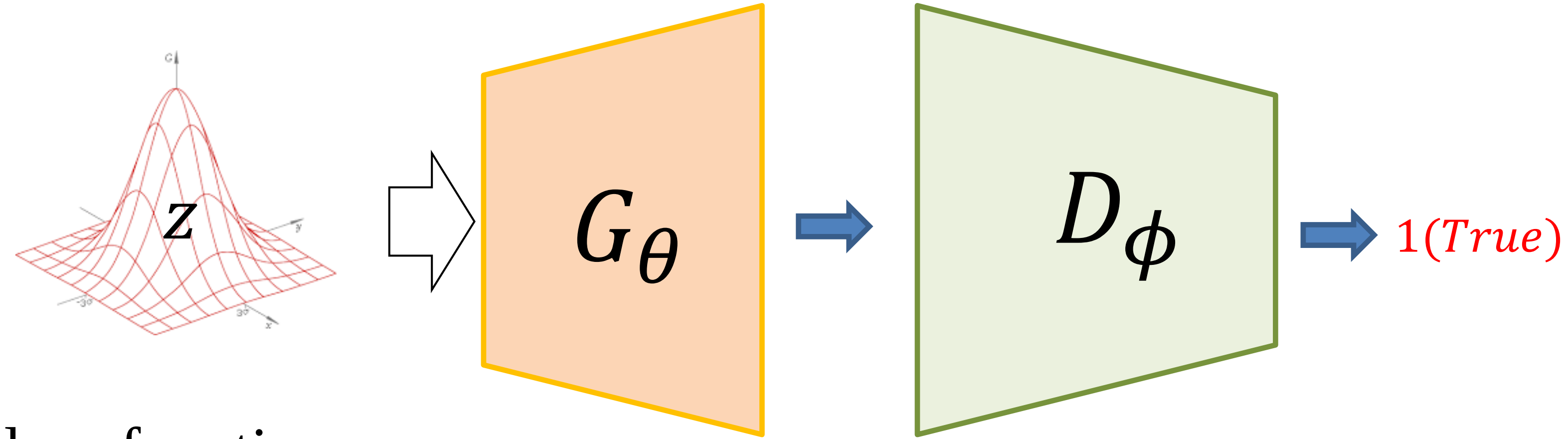

target loss function:

$$loss = E_{z\sim p(z)}[1 - D_\phi(G_\theta(z))]$$

Loss function of GAN

$G(z) \cong x \sim p_{target}(x)$

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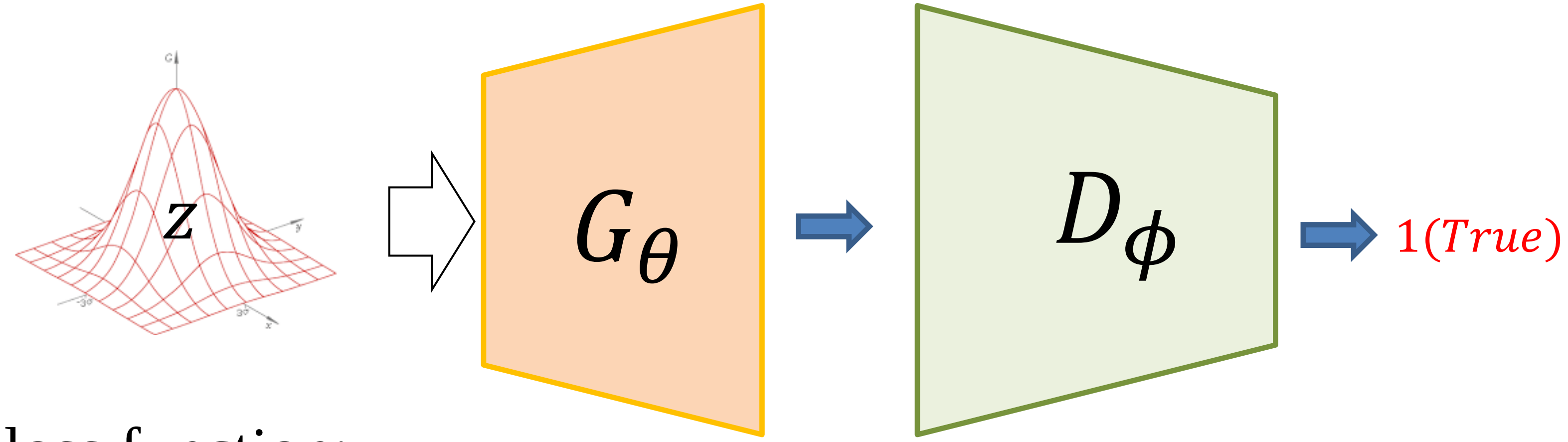

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$$loss = E_{z\sim p(z)}\left[\log\left(1 - D_\phi\left(G_\theta(z)\right)\right)\right]$$

Loss function of GAN

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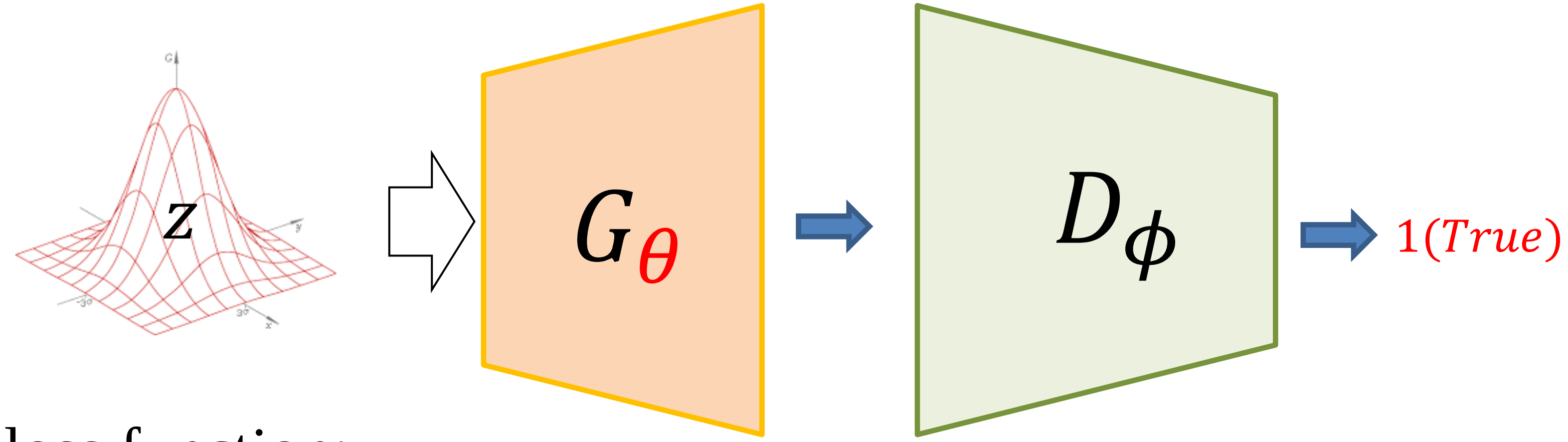


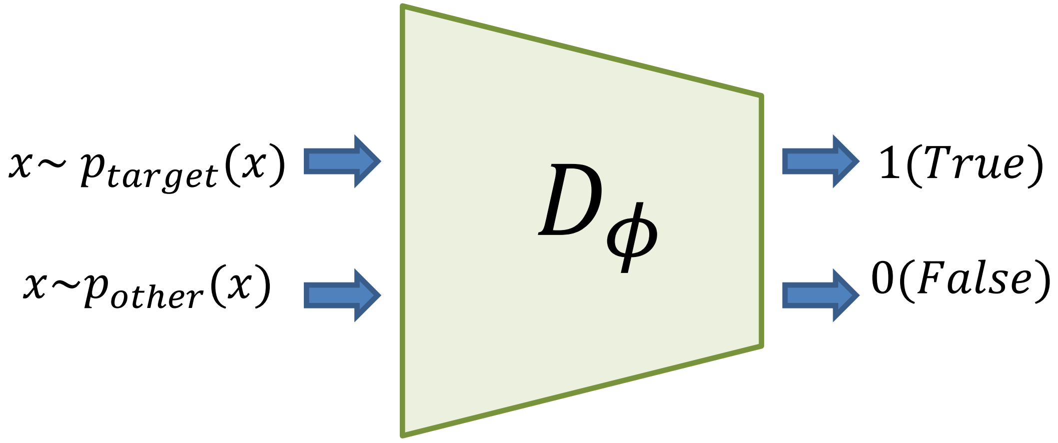
target loss function:

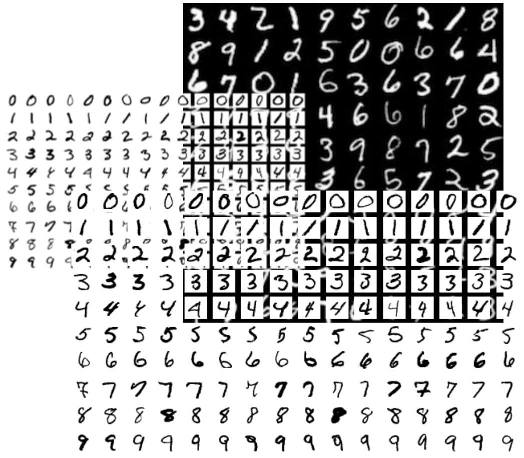
$$\underset{\theta}{argmin}\; E_{z\sim p(z)}\Big[\log\Big(1 - D_\phi\big(G_\theta(z)\big)\Big)\Big]$$

Loss function of GAN

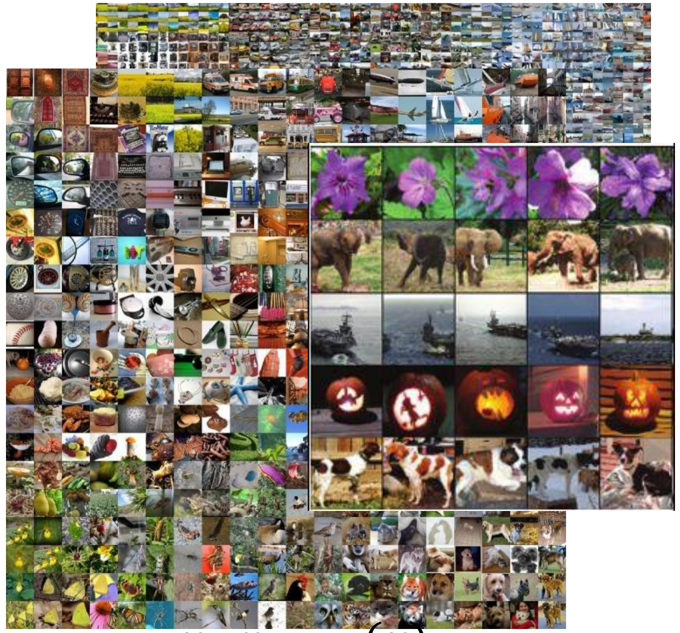
Domain Discriminator

1) in practice, it is impossible to collect all the data $x \sim p_{other}(x)$





$x \sim p_{target}(x)$

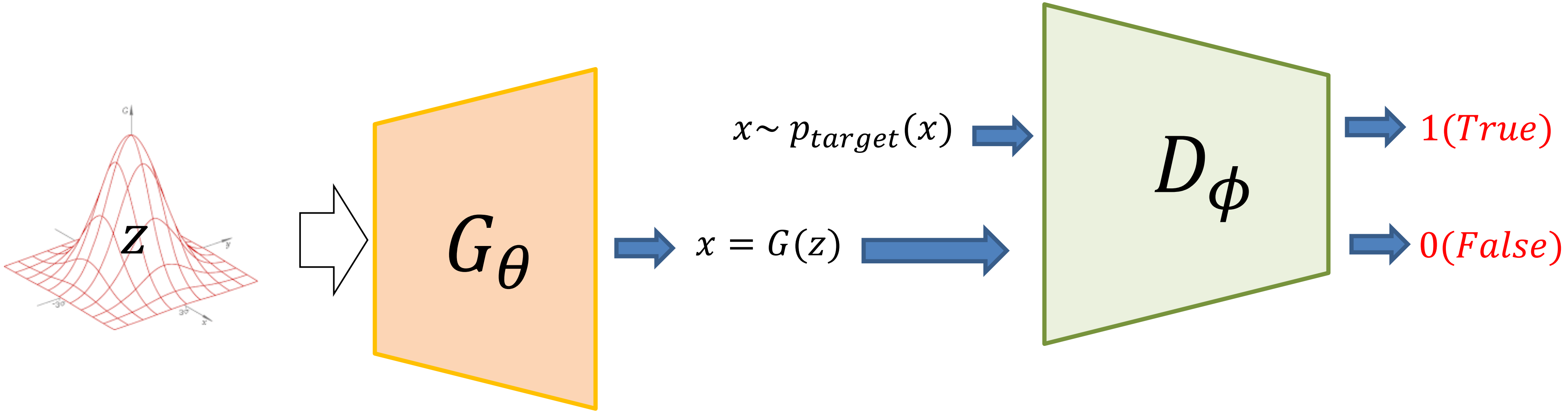


$x \sim p_{other}(x)$

2) our interest is only on $x \sim p(G(z))$ and $x \sim p_{target}(x)$

Loss function of GAN

Suppose that we already have G;
the ideal discriminator can recognize the difference
btwn $x \sim p_{target}(x)$ and $x = G(z)$



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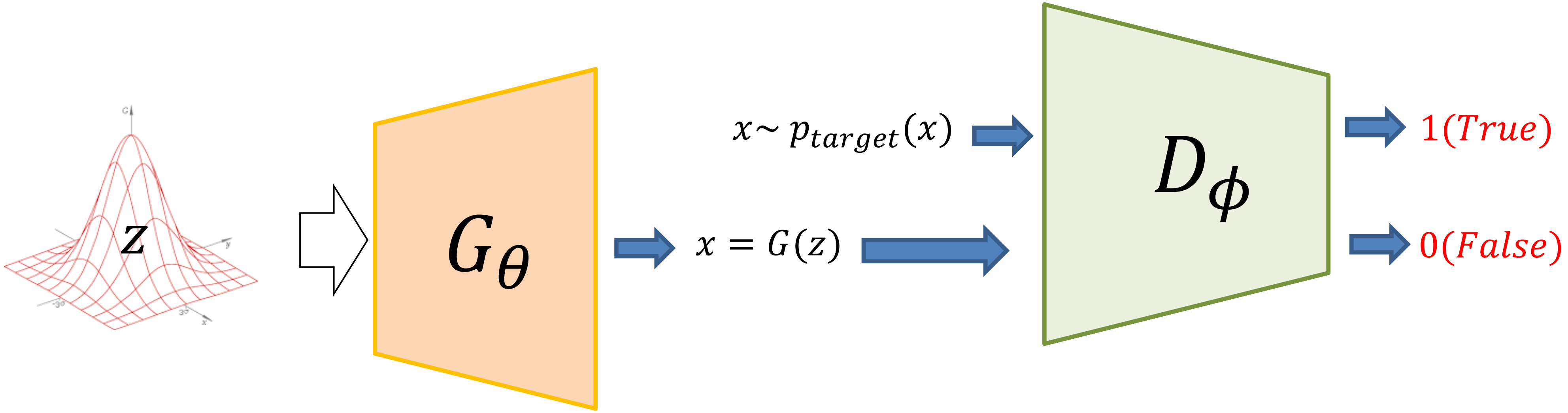

target loss function:

$$\underset{\phi}{argmax}\ E_{x \sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right] + E_{z \sim p(z)}\left[\log\left(1 - D_\phi\left(G_\theta(z)\right)\right)\right]$$

Loss function of GAN

Generator:

$$\underset{\theta}{argmin}\ E_{z\sim p(z)}\left[\log\left(1 - D_\phi(G_\theta(z))\right)\right]$$

Discriminator:

$$\underset{\phi}{argmax}\ E_{x\sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right] + E_{z\sim p(z)}\left[\log\left(1 - D_\phi(G_\theta(z))\right)\right]$$

Loss function of GAN

Generator:

$$\underset{\theta}{argmin}\ \boxed{E_{z\sim p(z)}\left[\log\left(1-D_\phi\left(G_\theta(z)\right)\right)\right]}$$

Discriminator:

$$\underset{\phi}{argmax}\ E_{x\sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right]+\boxed{E_{z\sim p(z)}\left[\log\left(1-D_\phi\left(G_\theta(z)\right)\right)\right]}$$

Loss function of GAN

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Discriminator:

$$\underset{\phi}{argmax}\ E_{x \sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right] + \boxed{E_{z \sim p(z)}\left[\log\left(1 - D_\phi\left(G_\theta(z)\right)\right)\right]}$$

total loss:

$$\min_{\theta}\ \max_{\phi}\ E_{x \sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right] + E_{z \sim p(z)}\left[\log\left(1 - D_\phi\left(G_\theta(z)\right)\right)\right]$$

$$V(G, D) = E_{x \sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right] + E_{z \sim p(z)}\left[\log\left(1 - D_\phi\left(G_\theta(z)\right)\right)\right]$$

Loss function of GAN

$$\min_{\theta} \max_{\phi} E_{x \sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right] + E_{z \sim p(z)}\left[\log\left(1 - D_\phi\left(G_\theta(z)\right)\right)\right]$$

$$= \min_{\theta} \max_{\phi} V(\theta, \phi)$$

$$V(G, D) = E_{x \sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right] + E_{z \sim p(z)}\left[\log\left(1 - D_\phi\left(G_\theta(z)\right)\right)\right]$$

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Global Optimum of GAN

existence and uniqueness of the global optimum

1) generator

- suppose that we have fixed $\theta(G)$; let's find optimal ϕ(D) (maximize V)

$$V(\theta,\phi) = \int_x p_{target}(x)\log(D_\phi(x))\ dx + \int_z p(z)\log(1 - D_\phi(\,G_\theta(z))\ dz$$

$$= \int_x p_{target}(x)\log(\,D_\phi(x)) + p_G(x)\log\,(1 - D_\phi(x))\ dx$$

where

$$p_G(x) = p(G_\theta(z))$$

Global Optimum of GAN

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$$V(\theta,\phi) = \int_x p_{target}(x)\log(D_\phi(x))\; dx + \int_z p(z)\log(1 - D_\phi(\,G_\theta(z))\; dz$$

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$$= \int_x \boxed{p_{target}(x)\log(\,D_\phi(x)) + p_G(x)\log\,(1 - D_\phi(x))}\,dx$$

$$\frac{\partial}{\partial y} a\log(y) + b\log(1-y) = \frac{a}{y} - \frac{b}{1-y} = 0$$

$$y = \frac{a}{a+b}$$

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$$\frac{\partial}{\partial y} a\log(y) + b\log(1-y) = \frac{a}{y} - \frac{b}{1-y} = 0$$

$$y = \frac{a}{a+b}$$

$$D_{\phi^*}(x) = \frac{p_{target}(x)}{p_{target}(x) + p_G(x)}$$

Global Optimum of GAN

existence and uniqueness of the global optimum

2) Discriminator

- we have ϕ^*; let's find optimal θ(G) (minimize V)

$$V(\theta,\phi^*) = E_{x\sim p_{target}(x)}\left[\log\frac{p_{target}(x)}{p_{target}(x)+p_G(x)}\right] + E_{x\sim p_G(x)}\left[\log\frac{p_G(x)}{p_{target}(x)+p_G(x)}\right]$$

$$= E_{x\sim p_{target}(x)}\left[\log\frac{p_{target}(x)}{\frac{p_{target}(x)+p_G(x)}{2}} - \log 2\right] + E_{x\sim p_G(x)}\left[\log\frac{p_G(x)}{\frac{p_{target}(x)+p_G(x)}{2}} - \log 2\right]$$

$$= E_{x\sim p_{target}(x)}\left[\log\frac{p_{target}(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right] + E_{x\sim p_G(x)}\left[\log\frac{p_G(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right] - \log 4$$

Global Optimum of GAN

existence and uniqueness of the global optimum

2) Discriminator

- we have θ^*; let's find optimal ϕ(G) (minimize V)

$$E_{x\sim p_{target}(x)}\left[\log\frac{p_{target}(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right]+E_{x\sim p_G(x)}\left[\log\frac{p_G(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right]-\log 4$$

$$=-\log 4+KL\left(p_{target}||\frac{p_{target}+p_G}{2}\right)+KL\left(p_G||\frac{p_{target}+p_G}{2}\right)$$

Global Optimum of GAN

existence and uniqueness of the global optimum

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$$E_{x\sim p_{target}(x)}\left[\log\frac{p_{target}(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right]+E_{x\sim p_G(x)}\left[\log\frac{p_G(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right]-\log 4$$

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Jensen-Shannon divergence (JSD)

$$\mathrm{JSD}(P||Q)=\frac{1}{2}(\mathrm{KL}(P||M)+\mathrm{KL}(Q||M)),M=\frac{1}{2}(P+Q)$$

Global Optimum of GAN

existence and uniqueness of the global optimum

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$$E_{x\sim p_{target}(x)}\left[\log\frac{p_{target}(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right]+E_{x\sim p_G(x)}\left[\log\frac{p_G(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right]-\log 4$$

$$=-\log 4+KL\left(p_{target}||\frac{p_{target}+p_G}{2}\right)+KL\left(p_G||\frac{p_{target}+p_G}{2}\right)\geq 0$$

Jensen-Shannon divergence (JSD)

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Global Optimum of GAN

existence and uniqueness of the global optimum

2) Discriminator

- we have θ^*; let's find optimal $\phi(G)$ (minimize V)

$$E_{x\sim p_{target}(x)}\left[\log\frac{p_{target}(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right]+E_{x\sim p_G(x)}\left[\log\frac{p_G(x)}{\frac{p_{target}(x)+p_G(x)}{2}}\right]-\log 4$$

$$= -\log 4 + KL\left(p_{target}||\frac{p_{target}+p_G}{2}\right)+KL\left(p_G||\frac{p_{target}+p_G}{2}\right)\boxed{\geq 0}$$

when $p_{target} = p_G$

$$G(z) \cong x\sim p_{target}(x)$$

$$p\big(G(z)\big) \cong p_{target}(x)$$

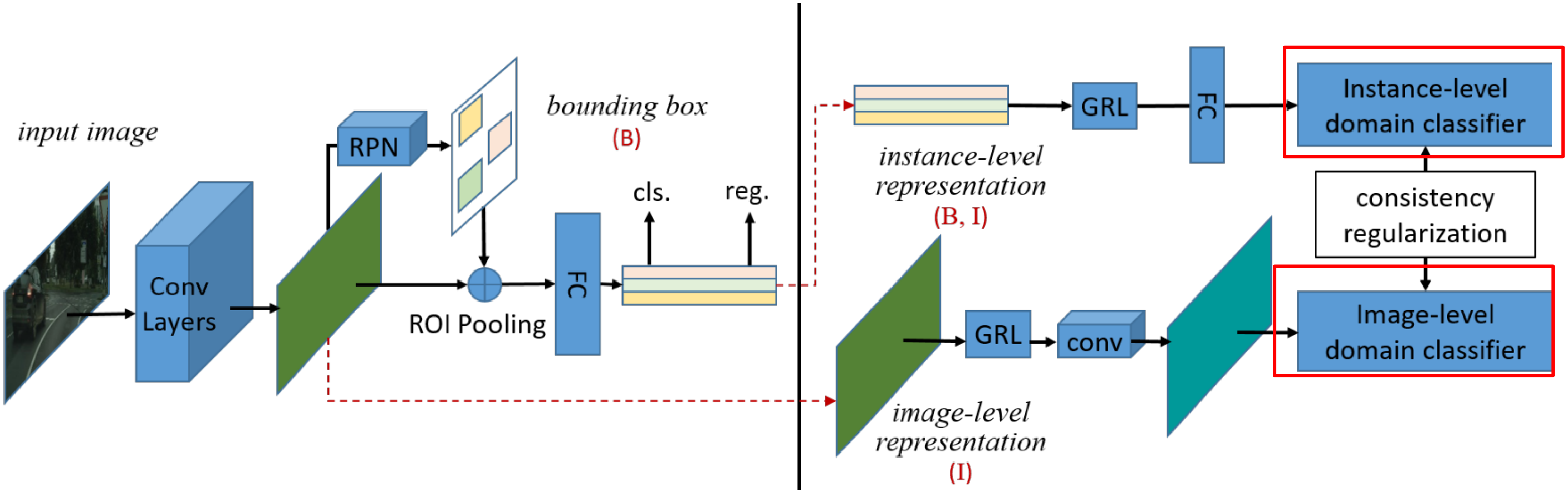
Jensen-Shannon divergence (JSD)

$$\mathrm{JSD}(P||Q)=\frac{1}{2}(\mathrm{KL}(P||M)+\mathrm{KL}(Q||M)), M=\frac{1}{2}(P+Q)$$

Examples of GAN

1) Cycle-GAN
2) Domain Adaptive Faster R-CNN for Object Detection in the Wild

Examples of GAN



(a) Faster R-CNN

(b) Domain adaptation components

Limitations of GAN

Limitations of GAN:

1) no probability distribution (no likelihood, no posterior for latent variables)

Limitations of GAN

Limitations of GAN:

1) no probability distribution (no likelihood, no posterior for latent variables)
2) highly dependent to Discriminator's performance

$$V(\theta,\phi) = E_{x\sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right] + E_{z\sim p(z)}\left[\log\left(1 - D_\phi(G_\theta(z))\right)\right]$$

Limitations of GAN

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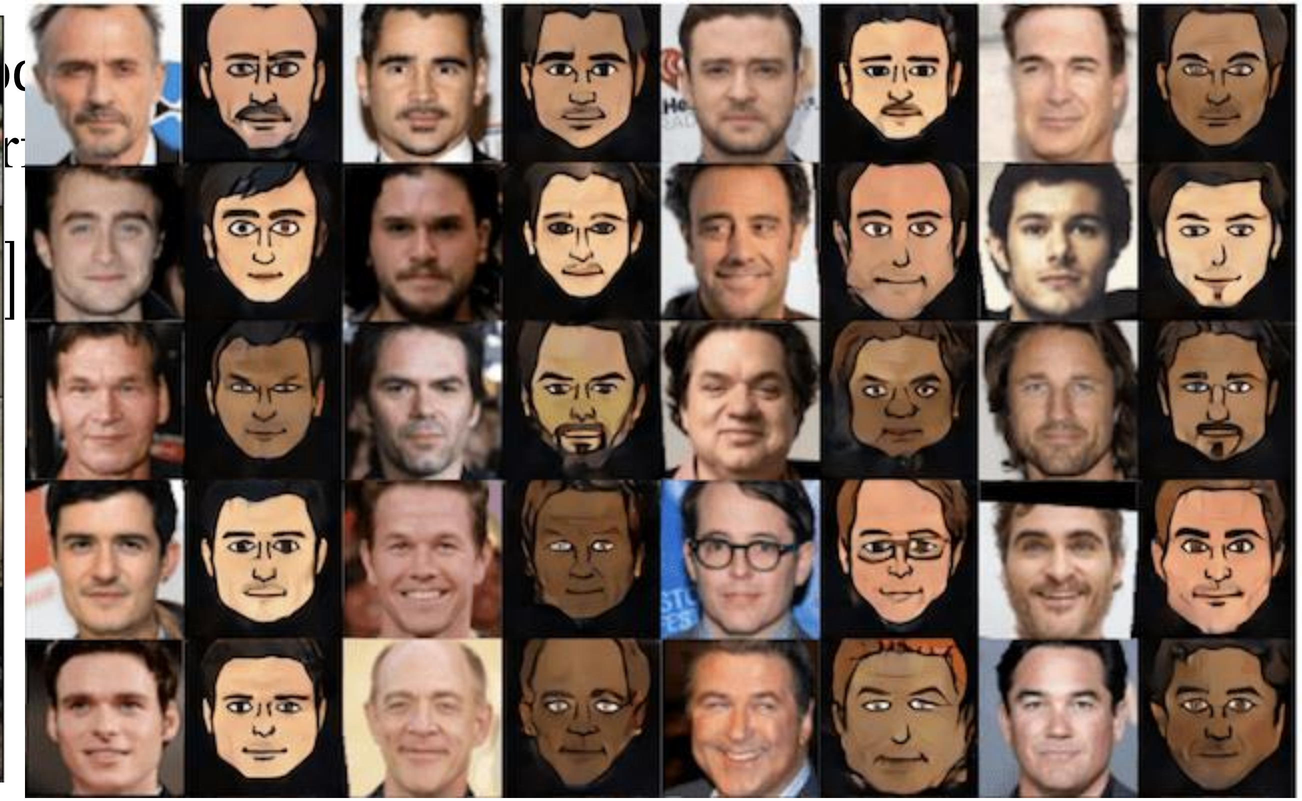
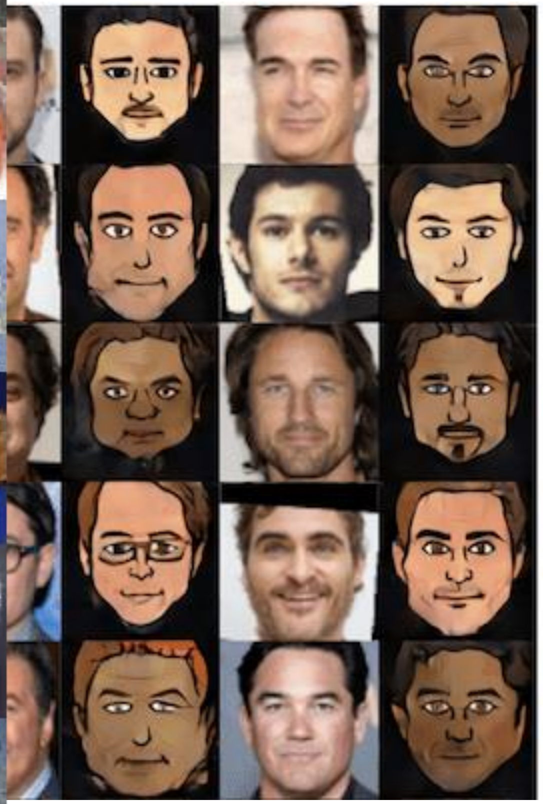


Limitations

Limitations of GA



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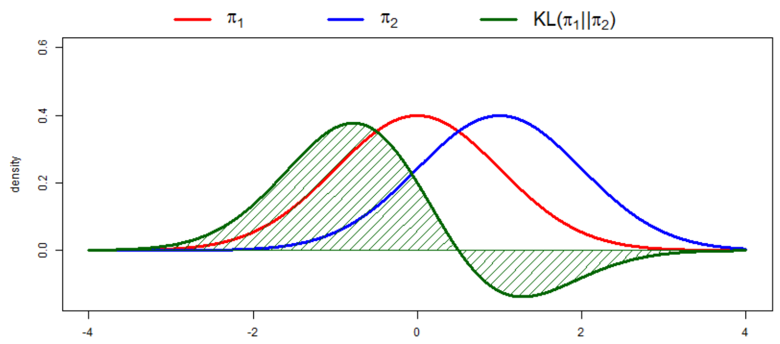
3) hard to train (converge) (why?)

$$-\log 4 + KL\left(p_{target}||\frac{p_{target}+p_G}{2}\right) + KL\left(p_G||\frac{p_{target}+p_G}{2}\right)$$



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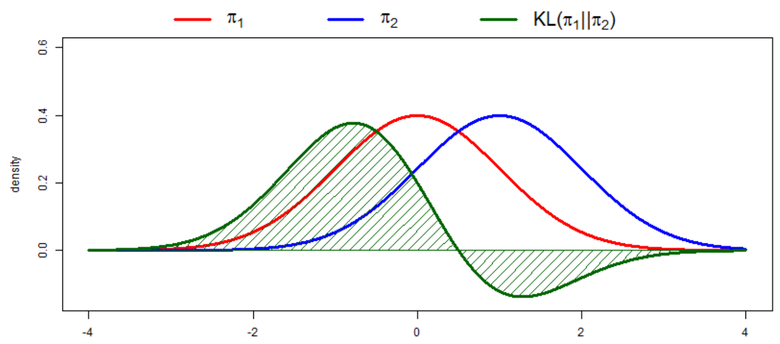
1) no probability distribution
2) highly dependent to Discriminator's performance

$$V(\theta,\phi) = E_{x\sim p_{target}(x)}\left[\log\left(D_\phi(x)\right)\right] + E_{z\sim p(z)}\left[\log\left(1 - D_\phi(G_\theta(z))\right)\right]$$

3) hard to train (converge) (why?)

$$-\log 4 + KL\left(p_{target}||\frac{p_{target}+p_G}{2}\right) + KL\left(p_G||\frac{p_{target}+p_G}{2}\right)$$



=> Wasserstein GAN, EB-GAN

Interim Summary

- Auto-encoding Variational Bayes
- Generative Adversarial Networks

Course Summary

Bayes Rule | Likelihood | Posteriori | Priori | Bayes Decision

Learning	Generative Model	Discriminative Model	Learning
ML(P)E, Bayes. L	GM	Lin. Classifier, SVM	LS, Convex O.
Histogram	K-NN, Parzen W.	Random Forest	Entropy
EM, MCMC	GMM	K-SVM, K-SVDD	Convex O.
MCMC, VI	Bayesian Net	Deep NN	SA, GA, BP(GD), NM
MLE, VI	Boltzm. Machine	ICA	K-L Divergence
MCMC, VI	Latent DA	Linear DA	Max. Separa.
EM	HMM	PCA	Max. Scatter

Course Summary